Jessie Chen

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Research Interests

Randomized Numerical Linear Algebra, Bayesian Optimal Sensor Placement, Optimization, Scientific Computing, Machine Learning

Education

North Carolina State University (GPA 4.0/4.0) — Raleigh, NC
PhD in Applied Mathematics — Aug 2022 – Present

- Co-advisors: Hangjie Ji and Arvind K. Saibaba
- Current Coursework: Data-Driven Modeling
- Previous Coursework: Uncertainty Quantification, Machine Learning in Biology, Matrix Methods in Data Science and Scientific Computing, Partial Differential Equations, Matrix Theory I/II, Numerical Analysis I/II, Mathematical Modeling I/II
- Qualifying Exams Passed: Matrix Theory, Numerical Analysis, Mathematical Modeling

North Carolina State University (GPA 4.0/4.0) — Raleigh, NC
MS in Applied Mathematics (En Route) — Aug 2022 – Dec 2024

Kennesaw State University (GPA 3.9/4.0) — Marietta, GA
BS in Computational and Applied Mathematics — Aug 2018 – May 2022

Honors and Scholarships

SHI Broader Engagement Participant at SIAM MDS 2024	Oct 2024
NSF RTG Research Fellowship	Aug 2023 – May 2025
NCSU DRUMS REU Research Assistantship	Summer 2023
Graduate School Teaching Assistantship	Aug 2022 – May 2023
AMS Undergraduate Student Travel Grant for JMM	Jan 2022
KSU URCA Fund Recipient	Fall 2020
President's List	Fall 2019 – Spring 2022
Dean's List	Fall 2018 – Spring 2019
HOPE Scholarship	Fall 2018 - Spring 2022

Research Experience

Sensor Selection for Gaussian Processes via Column Subset Selection
Mentor: Arvind Saibaba and Hangjie Ji (NCSU) — Jan 2024 - present

We study the optimal experimental design (OED) problem of sensor selection in Gaussian Processes with the perspective of column subset selection. We develop an algorithm which selects k out of n sensors to then reconstructs the spatial function using the selected sensor data. We also analyze lower bounds on the D-optimality of the sensor placements and explore Nyström approximations to further reduce computational complexity.

Accelerated and Weighted variants of Randomized Kaczmarz on Corrupted Linear Systems
Project Leads: Jocelyn Chi and Jamie Haddock — Aug 2025-Present

This project started at the Research Collaboration Workshop in Science of Data and Mathematics Conference (WiSDM) and is concerned with the acceleration of the Randomized Kaczmarz algorithm on corrupted linear systems. In addition, we propose various weighting schemes which aim to identify and remove the significance of the corrupted components in the linear system.

Direction of Arrival Estimation based on AI/ML Methods

Project Supervisors: Al Samuel, Rich Scholes, a nd Pranav Prabhakaran Sudha (Raytheon) Jun 2025

In partnership with representeatives from Raytheon and SIAM's Mathematical Problems in Industry (MPI) workshop, I worked in a team of graduate students and university professors to study the properties of AI/ML based approaches to solving direction of arrival estimation problems. My task in particular was to identify similarities between results from the AI/ML approach and the classical MUSIC algorithm approach by identifying patterns their associated matrices.

Randomized Sketch on Least Squares Problems

Mentor: Henry Boateng Jun 2025

As a part of SIAM's Graduate Student Mathematical Modeling Camp, my group consisting of other graduate students studied the effect of using non-conventional sketch matrices for randomized least squares problems in a computationally efficient manner. Specifically, we look at non-Gaussian sketch matrices such as the Sparse Johnson-Lindestrauss Transform and the Subsampled Randomized Hadamard Transformation and employ them in the Randomized SVD algorithm.

Deep Learing for Cryo- ET Images

Mentor: Noushin Hajarolasvadi (Zuse Institute Berlin and Thermo-Fisher Scientific) Jun 2024 – Aug 2024

This project was organized through the Institute for Pure and Applied Mathematics (IPAM) under the Graduate Research in Industrial Problems (G-RIPS) program. We used deep learning methods on the tasks of orientation and orientation estimation of macromolecules captured by Cryo-ET imaging. We also explore explainable AI (XAI) methods to further understand decisions made by our designed networks.

Dynamics of Liquid Metal Droplets with Electrochemical Oxidation

Mentor: Hangjie Ji (North Carolina State University) May 2023 – Jan 2024

We develop a 1D model for liquid metal droplets sliding down an incline plane whose surface tension changes via electrochemical oxidation. We also compare numerical simulation results to experimental results from the Daniels Lab at NC State. In this REU project, I served as a graduate research assistant.

Optimizing Radiotherapy Treatments of Brain Metastases

Mentor: Dávid Papp (North Carolina State University) May 2021 – Aug 2022
In a team with three other undergraduate students as part of an REU, we investigated the computational effects of an alternative approach to treating brain metastases. Since each patient is unique, a new optimization model must be introduced within a reasonable computation time. We presented our research in the AMS-PME Undergraduate Poster Session at JMM 2022.

Analysis of Congressional District Maps in Georgia
Mentor: Andrew Wilson (Kennesaw State University) Jan 2021 – May 2022
The purpose of this undergraduate research is to bring a probabilistic and graph theoretic mindset towards redistricting to GA policymakers and voters. We use various computational methods and algorithms to analyze the effects of a given proposed map. We presented our findings publicly at Posters at the Georgia Capitol to help voters get a sense of fairness in proposed maps so that democracy is promoted and gerrymandering is prevented.

Talks, Tutorials, and Posters

SIAM CSE 2025 Minisymposium Mar 2025
SIAM MDS 2024 Poster Session Oct 2024
AMS/SIAM NCSU MATLAB Tutorial Nov 2023
REU MATLAB Tutorial Jun 2023
JMM AMS-PME Student Poster Session Apr 2022
Posters at the Georgia Capitol Mar 2022
KSU Symposium of Student Scholars Presentation Nov 2021
KSU Research Spotlight Talk Oct 2021
NSCU Summer Undergraduate Research Symposium Jul 2021

Campus Involvement

AWM NSCU Comittee Member Oct 2023 - Aug 2024
SIAM NCSU Student Chapter President Aug 2023 – Present
Dean's Scholars Council Member Jan 2021 – May 2022
Learning Assistant Jan 2019 – May 2021
Courses assisted: Calculus I, Calculus II, Ordinary Differential Equations
Math Club President Fall 2019 – May 2022
Student Advisory Board Member Fall 2019 – Spring 2020

Memberships

Society for Industrial and Applied Mathematics July 2022
Pi Mu Epsilon Apr 2022
Math Alliance Scholar Aug 2020

Conferences Attended

SIAM Uncertainty Quantification Mar 2026
SIAM Computational Science and Engineering Mar 2025
SIAM Mathematics of Data Science Oct 2024
Joint Mathematics Meeting 2024 Jan 2024
Joint Mathematics Meeting 2023 Jan 2023
Joint Mathematics Meeting 2022 April 2022

KSU Symposium of Student Scholars Nov 2021
NCUWM Conference Jan 2021
Field of Dreams Conference Nov 2020

Programming Skills

Proficient in MATLAB
Experience in Python
Limited Experience in Java, SAS, R, Mathematica